

The World-Model Specification:

an invariance-forced theory of latent dynamics

V1 — internal draft, machine-checked corpus `WMSpec`

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Abstract

We present a theory of what a world model *is*, derived by closure of an impossibility family rather than by stipulation. The theory has five levels — encoder, objective, dynamics, metric, and architecture — each carrying characterization theorems (both directions), each instantiated on literal formalized physics, and each equipped with executable witnesses whose empirical antecedents are computed inside the proof kernel. Every result below is machine-checked in Lean 4 over a pinned Mathlib, axiom-clean (`propext/Classical.choice/Quot.sound`; several core results use no axioms at all), with no `sorry` and no `native.decide`. The corpus is `design-lab/world-models/WMSpec` (~20 modules); the running two-axis research ledger is `world-models/URS.md`.

The headline: “world model” is not a property of a network. It is a three-place relation between a representation, a declared target family, and a readout budget — and once the relation is stated, the mathematics forces everything else: which readouts can exist (a McShane characterization), which objectives can see which structure (an invariance calculus with exact separation laws), which state abstractions survive (a forcing theorem: the bisimulation quotient), what geometry the quotient carries (Fisher–Rao, with a data-processing inequality), and which architectures are possible for which physics (an admissible-horizon subgroup law with a three-point spectrum). The strong-form conclusion for JEPA-class systems is stated in §9.

1 The thesis

A representation $E : S \rightarrow Z$ of a dynamical environment is called a world model in the literature on evidence that cannot support the phrase. The theory below replaces the phrase with a specification, proves that the specification is *forced* (any weaker reading is refuted by a machine-checked counterexample; any reading at least as strong collapses onto ours), and then develops the positive structure the specification generates.

Definition 1.1 (The specification, informal). *Fix a target family $\mathcal{T} \subseteq (S \rightarrow V)$ (the quantities the model is for) and a readout budget (a Lipschitz constant L , or a readout class). A representation E is a world model relative to $(\mathcal{T}, L)$ when:*

- (i) **Separation:** $\ker E$ refines every $T \in \mathcal{T}$ — states the targets distinguish, the encoder distinguishes;
- (ii) **Metric domination:** $|Ts_1 - Ts_2| \leq L \cdot d_Z(Es_1, Es_2)$ — latent geometry prices target gaps;
- (iii) **Congruence:** $\ker E$ is a congruence for the dynamics — a latent transition law exists at all.

Each clause is the contrapositive of a proved impossibility, and two of the three clauses admit *characterizations* (if-and-only-if theorems), so the specification is not a wish list: it is exactly the boundary of the refutable.

Principle. “World model” is a three-place relation. A claim of the form “ E is a world model” without a declared $\mathcal{T}$ and readout budget is not false; it is unevaluable. Every theorem below is relative to declared $(\mathcal{T}, L)$, and several results show how the truth value changes as the declaration changes.

2 The negative layer

2.1 The master lemma and its orbit form

Lemma 2.1 (`not_factorThrough_of_collapse`; *no axioms*). *If $Es_1 = Es_2$ and $Ts_1 \neq Ts_2$ then no $g : Z \rightarrow W$ satisfies $T = g \circ E$.*

Lemma 2.2 (`not_factorThrough_of_invariance`; *no axioms*). *If $E \circ b = E$ pointwise (the encoder declares b nuisance) and $Ta \neq T(ba)$ (the target carries b as signal), no readout of E recovers T .*

The orbit form is the symmetry-native negative layer: mismatched invariance — an encoder invariant under a transformation the target moves — is fatal for every readout, of any regularity. The quantitative version (`approx_lipschitz_ineq`) replaces exact collapse with ε -proximity and bounds every L -Lipschitz readout by $|Ts_1 - Ts_2| \leq L d(Es_1, Es_2)$.

2.2 Literal physics instances

All physics below is PhysLean’s, not hand-rolled: the harmonic oscillator structure, its energy functional, the proven conservation law `energy_conservation_of_equationOfMotion`, and the closed-form solution flow.

Theorem 2.3 (`energy_not_recoverable_from_position`). *For every position x_0 and nonzero velocity v , the position-only encoder collapses the pair $\langle x_0, 0 \rangle, \langle x_0, v \rangle$ while the conserved energy separates it; no readout of the position latent computes the energy of the evolution.*

Theorem 2.4 (`worldModel_latent_geometry_bound`). *Any encoder supporting an L -Lipschitz energy readout obeys $\frac{1}{2}m\|v\|^2 \leq L \cdot d(E\langle x_0, 0 \rangle, E\langle x_0, v \rangle)$: literal physics prices the latent geometry of every faithful representation. At L fixed and $v \neq 0$ the fiber-collapse impossibility is the degenerate case.*

Theorem 2.5 (`positionLatent_dynamics_iff_stroboscopic`). *For the literal oscillator flow, a latent transition map for the position-only encoder exists at horizon t iff $\sin(\omega t) = 0$. Away from the half-period lattice the position latent admits no dynamics law at all; on it, the flow aliases to $\cos(\omega t)x_0$ and the latent step exists by construction. What looks like a world model at stroboscopic sampling times is aliasing.*

The self-grading theorem (`latentDistanceSuccess_not_true_state_functional`, in the companion `LewmValidity` package) closes the loop to evaluation practice: a latent-metric success criterion is not a function of true-state attainment once the encoder sheds a goal-relevant direction — the structural form of the empirical planner dissociation (latent success ≈ 0.98 , true-goal correlation ≈ 0.30).

3 The characterization suite

3.1 Readability is exactly metric domination

Theorem 3.1 (McShane characterization; `lipschitz_readout_iff`). *$\exists g$ L -Lipschitz with $T = g \circ E$ iff $|Ts_1 - Ts_2| \leq L d(Es_1, Es_2)$ for all pairs. Necessity is the approximate-Lipschitz bound; sufficiency pushes T along E (well-defined by domination-at-zero) and extends by McShane (`LipschitzOnWith.extend_real`).*

Corollary 3.2 (`no_lipschitz_readout_iff`). *No L -Lipschitz readout exists iff some pair’s target gap strictly exceeds L times its latent gap: the quantitative obstruction is exactly equivalent to unreadability.*

This converts specification clause (ii) from a necessary condition into an exact criterion at every budget L , and resolves the standing confusion between “information present” (unrestricted readouts) and “planner-usable” (bounded L): they are the same specification evaluated at different budgets.

3.2 The objective layer: an invariance calculus

An objective family $L : \Theta \rightarrow X \rightarrow M \rightarrow V$ has an invariance class `ObjectiveInvariant` $Lg \iff \forall \theta x m, L\theta(gx)m = L\theta xm$, organized as a submonoid of `End(X)` (`invarianceMonoid`); the mask-indexed refinement $G_L(m)$ (`invarianceMonoidAt`) satisfies $G_L = \sqcap_m G_L(m)$ (`invarianceMonoid_eq_iInf`). The calculus’s laws:

- **Blindness** (`objective_blindness`, `Quot.sound` only; `not_factorThrough_loss_of_invariance`, *no axioms*): an invariant g makes every loss observable factor through g ; a target moving along a g -orbit is recoverable from *no* observable of the objective.
- **Zero power** (`blind_tests_zero_power`): for $b \in G_L$, every Boolean test built from a loss observable reads *identical* expectations on P and on the shifted law $b_{\#}P$ — even when the shift is real. With the Fisher–Rao bound below, the power spectrum is bracketed: general tests $\leq \sin(d_{\text{FR}}/2)$, loss-blind tests = 0.
- **Mask strata**: the empty mask constrains nothing (`objectiveInvariantAt_nil_of_additive`); singleton masks generate (`objectiveInvariant_of_singletons`) — for mask-additive objectives, all-mask invariance already forces componentwise invariance, a *structural* cancellation exclusion.
- **The invisibility instance**: the verified JEPA loss algebra (`jepaLoss_target_ext`) is precisely a mask-indexed invariance statement: rewriting unselected targets lies in $G_L(I)$. “The mask chooses what the objective can see” is a theorem about how $G_L(m)$ varies with m .

3.3 Matched objectives: separation laws with hypotheses in minimal locations

For the repaired matched form $L_{\text{matched}} = \iota(L_{\text{pred}}) + \sum_j \lambda_j G_j$ (data-reading guards; the verified contract’s constant guard slot is provably invariance-blind, `constant_guard_invariant`):

Theorem 3.3 (`matched_invariance_iff`). *For a mask-additive predictive branch, $g \in G_{L_{\text{matched}}}$ iff the predictive branch is g -invariant and the guard sum is g -invariant. Both directions hold with no richness hypothesis: the empty mask zeroes the predictive branch, separating the joints unconditionally.*

Theorem 3.4 (`guards_invariant_of_matched`, `matched_invariance_characterization`). *Positive weights and monotone guard deviations — needed only at the guard/guard joint — split the invariant guard sum into per-guard invariance, giving the full $G_{L_{\text{matched}}} = G_{\text{pred}} \cap \bigcap_j G_{G_j}$, bound to the literal `jepaLoss` via its additivity.*

Theorem 3.5 (The exchange constant; `mixed_guard_exchange`). *For a mixed objective with data guards and law guards under the joint action (g, h) , invariance forces exactly a one-dimensional cancellation channel: a constant c with data-deviation sum $\equiv c$ and law-deviation sum $\equiv -c$, uniformly. Monotone deviations on both families collapse $c = 0$ and full separation follows (`mixed_invariance_characterization`); the pushforward-coupled instance ($h = \text{map}_g$ on finite laws) is the formal home of “guards computed on the representation distribution.”*

Guard families are unified as *characterized guards* (`CharacterizedGuard`: value, property, zero-iff) with one transport theorem — “the guard enforces the symmetry it names” — proved once for every family. HSIC is characterized unconditionally given a characteristic kernel; mutual information only under per-point side conditions (absolute continuity, finite KL) at *both* points — a scope law for guard *choice*.

4 The forcing theorem

On finite S with probabilistic dynamics $P : S \rightarrow \text{FintypePMF } S$ and target family $\mathcal{T}$, call a setoid a *spec congruence* when it refines every target and gives equal mass to every saturated set (the congruence is stated *quotient-free*; classes appear only as theorems).

Theorem 4.1 (The forcing theorem; `bisim_mem`, `forcing_theorem`). *The spec congruences have a greatest element — bisimilarity, their lattice supremum, which is itself a spec congruence (Larsen–Skou union-of-bisimulations in finite form: one `EqvGen` chain induction carried by a saturation lemma). Every spec-satisfying encoder’s kernel lies below it.*

Theorem 4.2 (Universal factorization; `bisim_classification_factors`). *The bisimulation classification factors through every spec-satisfying representation: whatever latent space a world model uses, it already computes the causal states.*

Theorem 4.3 (The quotient satisfies the specification; `quotientTarget`, `quotientDynamics`, `quotientDynamics_mk`). *Targets descend to the quotient; the dynamics descends as the pushforward of the step law along the quotient map, with well-definedness being the congruence property and the commuting square holding by `rfl`.*

This is the apex of the positive theory: the specification does not merely admit the bisimulation quotient — it forces it. Identifiability-style positive results (LeJEPa’s linear identifiability under Gaussian isotropic exploration) are regime-bound sufficient routes *inside* this envelope; decodability-on-frame-splits and latent-distance success fail the specification’s clauses outright (machine-checked witnesses); what survives the closure is exactly the causal-state/bisimulation structure.

5 The metric layer

The forced quotient carries a canonical geometry: $\text{bisimMetric}(c, c') := d_{\text{FR}}(P_q c, P_q c')$, the Fisher–Rao distance between one-step class laws (well-definedness inherited from the forcing theorem — the metric exists *because* P3 holds).

Theorem 5.1 (Pricing; `bisimMetric_prices_tests`, `bisim_zero_one_step_power`). *Any Boolean class test’s one-step expectation gap between classes is at most $\sin(\text{bisimMetric}/2)$; bisimilar states have identical one-step test power.*

Theorem 5.2 (Data-processing inequality; `fisherRao_mapPMF_le`). *Coarse-graining never increases Fisher–Rao distance: the Bhattacharyya coefficient increases under every pushforward (per-fiber Cauchy–Schwarz) and arccos is antitone. The finite Chentsov-monotonicity brick: quotienting never inflates distinguishability.*

Supporting this layer: the finite Fisher–Rao risk bound (`finite_fisherRao_risk_bound`) at the *tight* constant — $|E_p f - E_q f| \leq \sin(d_{\text{FR}}(p, q)/2)$ — via a tight transfer lemma ($\text{gap} \leq \text{tv}/2$, the $\sum(p - q) = 0$ complement argument), which also surfaced a convention finding: the existing verified transfer hook is ℓ^1 -typed, twice standard TV; composing it naively loses a factor of 2.

Conjecture 5.3 (Gated, ChallengeCrown form). *(i) The Ferns-style discounted fixpoint: bisimMetric is the one-step layer of the unique fixed point $d = \max(\text{target gap}, \gamma W_1(d)(P_q, P_q))$ (needs the finite Wasserstein layer). (ii) Chentsov uniqueness: any class-metric natural under congruent embeddings and monotone under pushforward agrees with a rescaling of Fisher–Rao. Together: the specification forces the partition and its geometry.*

6 Flow architectures and the spectrum

Flows are additive actions of time on states; an encoder admits a latent step at horizon t iff $E \circ \Phi_t$ factors through E (`latentStep_iff_factorsThrough` — the oscillator dichotomy is an instance).

Theorem 6.1 (The horizon laws; `admissibleHorizons`, `admissibleSubgroup`). *The admissible horizons of any encoder against any flow form an additive submonoid of time (no axioms). For finite state spaces and group time they form a subgroup: a pigeonhole period $k \cdot t \equiv \text{id}$ gives $(-t) +^v x = ((k-1) \cdot t) +^v x$, so negative horizons ride the forward ones.*

Theorem 6.2 (Necessity of finiteness; `one_mem_collapse`, `neg_one_not_mem_collapse`). *On $\mathbb{Z}$ with the collapse-nonnegatives encoder, $+1$ is admissible and -1 is not: the admissible set is a genuine submonoid-not-subgroup.*

Theorem 6.3 (The spectrum). *Three physics classes, one law:*

$$\underbrace{\{0\}}_{\text{kinematics (kinematic_admissible_iff)}} \subset \underbrace{(\pi/\omega)\mathbb{Z}}_{\text{oscillator (oscillator_latentStep_iff_int_multiple)}} \subset \underbrace{\text{all of } T}_{\text{quotient-homomorphic encoders}}$$

Kinematic aliasing never rescues a velocity-shedding latent; the oscillator’s admissible set is the half-period lattice on PhysLean’s literal flow; group-quotient encoders alias always.

The architecture is a type: `FlowWorldModel` packages an encoder, a *declared* horizon submonoid, and a step whose `commutes` field is the flow-correctness certificate; admissibility of the declared horizons is *derived* (no axioms), so the certificate cannot be circular. The executable warmup (`blockModel`: shift flow on $\mathbb{Z}/6$, two-block encoder — the discrete shadow of half-period aliasing) has its dichotomy computed by kernel `decide` and its step certified by closure induction. Named obligation: the oscillator’s `AddAction` packaging awaits the flow group law (closed-form trigonometric route recorded).

7 The executable layer

Executed float32 values are dyadic rationals; the corpus adds the computable $\mathbb{Q}$ -reading (`dyadicToRat`, `toRat`) with bridges proving cast-agreement with the noncomputable real semantics. Consequences: bit-level decodes by kernel `decide` (choice-free: `[propext, Quot.sound]`), executed values as exact rationals (`toReal_ofBits_ulp` = $1 + 2^{-23}$), and witness obligations discharged by computation-plus-cast rather than analysis. The executed one-ULP witness shows a collapse-blind predictive branch (loss `rfl`-computes to zero on both sides) while the executable spread guard detects the collapse on literal bits (`witness_executedGuard_matched_detects`); the n -point variance guard is Tensor-typed on the literal TorchLean carrier with its own zero-iff. Two witness tiers coexist inside one objective: executable guards `decide`; analytic guards transport as Prop-iffs — the journal’s tier split manifesting per-guard.

8 The novel mathematics, acutely

Genuinely new as stated (no informal source known to us; several were conjectured inside the research ledger before any proof existed and were born verified):

1. **The exchange constant** (Theorem 3.5): between two guard families under a joint action, the cancellation space is exactly one-dimensional. Nothing weaker (free separation) and nothing stronger (arbitrary cancellation) is true.
2. **The admissible-horizon laws**: submonoid always; subgroup under finiteness via the pigeonhole-period argument; necessity of finiteness by explicit counterexample; and the three-point spectrum classifying physics by the algebra of its aliasing set.
3. **The nil-mask separation**: the vacuity stratum of the mask lattice (empty mask constrains nothing) inverted into an unconditional separation tool at the predictive/guard joint — eliminating the “independent components” hypothesis from the matched-objective characterization at that joint and localizing all richness at the guard/guard joint.
4. **The quotient-free congruence formulation** enabling a one-induction Larsen–Skou: equal mass on saturated sets replaces class talk, so bisimilarity’s definition never mentions its own classes and membership-of-the-supremum is a single `EqvGen` induction.
5. **The stroboscopic dichotomy as a theorem about literal formalized physics**, and its kinematic anti-thesis ($\{0\}$ exactly).

New formalizations of known mathematics (first machine-checked forms, to our knowledge): the McShane readability characterization as a world-model theorem; finite Larsen–Skou; the finite Fisher–Rao risk bound at the tight constant; the Fisher–Rao data-processing inequality by per-fiber Cauchy–Schwarz; the executed-dyadic $\mathbb{Q}$ -reading bridge. Known results consumed as anchors: PhysLean’s conservation laws and solution flow; the verified JEPA/ predictive-view algebra; Mathlib’s setoid lattice and Lipschitz extension.

9 The strong-form conclusion on JEPA-class systems

Internal statement; the public form remains the scoped route-closure. With that understood:

JEPA, as constituted, is a predictive-quotient learner whose objective provably cannot see the structure the phrase “world model” promises, whose evaluation practices provably cannot detect the failure, and whose one proven identifiability regime is the regime in which a world model is unnecessary. Specifically, each clause machine-checked: (a) the loss is extensional over selected targets — structure outside the mask, the predictor, and the representation loss contributes nothing (`jepaLoss_target_ext`), and the geometry-guard slot of the verified contract is a constant, hence invariance-blind (`constant_guard_invariant`); (b) every transformation in the objective’s invariance monoid is invisible to every loss observable and to every loss-built test on shifted laws — zero power, exactly (`blind_tests_zero_power`); (c) in the deployed, goal-directed regime, the learned latents demonstrably shed task-relevant directions, and then: no readout recovers the shed quantities (energy included, against a *proven* conservation law), no latent dynamics law exists at generic horizons, and latent-metric success self-grades — 0.98 reported success at 0.30 true-goal correlation is not a bug but the predicted signature; (d) the identifiability theorems that would rescue the phrase hold under Gaussian isotropic exploration and exact regularization — their own authors concede goal-directed deployment exits the regime. The honest composite: *whatever JEPA training produces, the evidence offered for calling it a world model is compatible, to the level of machine-checked equivalence, with a system that has memorized the mask algebra and nothing else; and the specification any genuine world model must meet — forced, not chosen — is bisimulation structure that the JEPA objective does not optimize and its guard slot cannot express.*

Rails that survive even the strong form: the middle arrow (trained encoder inherits the objective’s invariance) is an architectural hypothesis, never a theorem of training dynamics; passing grouped-split probes (Joseph et al.) is genuine anti-collapse evidence for the probed quantity on the probed distribution; and matched objectives — prediction plus one data-reading guard per broken symmetry, weights positive, deviations monotone — are exactly the repair this theory licenses, with the guard laws (§3.3) saying which guards can enforce what.

10 Status ledger

Kernel-checked and axiom-clean: the entire `WMSpec` corpus (negative layer, characterization suite, forcing theorem, metric bricks, flow stratum, executable layer), the `LewmValidity` companion (12 audited headlines incl. the self-grading theorem), the non-identifiability modules in `TLT_Proofs`. No-axiom results: master lemma, orbit lemma, blindness bridge, submonoid law, architecture-admissibility, mask-intersection iff. Gated (ChallengeCrown): Ferns fixpoint; Chentsov uniqueness; the oscillator `AddAction` group law; quantum stratum (observable-subalgebra encoders, Heisenberg invariance — the next mini-project’s crown). The research ledger (`URS.md`) records, for every stratum, the pre-registered conjectures, sophism registries, predicted-vs-fired oscillations, and the banked proof-engineering rules.